

3ML3 - Exercise 4

we start with: $g(b, w) = \frac{1}{p} \sum_{p=1}^p (b + f_p^T w - y_p)^2 + \lambda \|w\|^2$

we can write: $w = Fz + r$, where $F^T r = 0_{p \times 1}$

Put in model: $f_p^T w = f_p^T Fz$

$H = F^T F$, $h_p = F^T f_p$, so $f_p^T Fz = h_p^T z$

This gives us: $\frac{1}{p} \sum (b + h_p^T z - y_p)^2$

$\|w\|^2 = \|Fz + r\|^2 \Rightarrow \|w\|^2 = \|Fz\|^2 + \|r\|^2 = \|Fz\|^2$

$\|Fz\|^2 = z^T F^T F z = z^T H z$

This results in: $g(b, w) = \frac{1}{p} \sum_{p=1}^p (b + h_p^T z - y_p)^2 + \lambda z^T H z$

and the kernelized model: $\text{model}(x_p) = b + h_p^T z$